

JIYA YADAV

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in production agentic systems, fine-tuned transformer models, and retrieval-augmented generation (RAG). Architected sub-500ms multi-agent backend pipelines with FastAPI, PyTorch, and Groq LLM streaming. Experienced in implementing hybrid explainable AI (SHAP & lexicon attributions), multi-signal crisis guardrails, and full observability stacks (Prometheus, structlog). Demonstrated success fine-tuning DistilRoBERTa models achieving 0.89 macro F1 with 96% recall on crisis detection.

TECHNICAL SKILLS

- AI/NLP & Agentic Systems: PyTorch, Hugging Face Transformers, Fine-Tuning (DistilRoBERTa), Multi-Agent Orchestration, RAG, Groq API (groq/compound-mini), Sentence-Transformers (all-MiniLM-L6-v2), SHAP.
- ML & Backend Engineering: FastAPI, AsyncIO, Streamlit, Pydantic v2, ChromaDB (Vector Store), Scikit-learn, XGBoost, REST API Design.
- MLOps & Observability: Prometheus Metrics, structlog (JSON Logging), Git, GitHub Actions (CI/CD), pytest, Docker, Hugging Face Hub/Spaces, Ruff, Codecov.
- Data & Cloud Analytics: Python (Advanced), SQL, PostgreSQL, Apache Airflow, Pandas, NumPy, Power BI, DAX, Google Colab (T4 GPU).

FEATURED PROJECT

Mental Health Agentic AI Platform | [Live Demo](#) | [GitHub Source](#) | [HF Model](#) |

Production-grade multi-agent NLP platform for emotion intelligence, vector RAG, explainable AI, and crisis guardrails.

- Fine-Tuning & Model Deployment: Fine-tuned DistilRoBERTa on 16K emotion samples using AdamW + linear warmup on Google Colab T4 GPU, boosting macro F1 from 0.87 to 0.89 with 96% recall on crisis signals. Published model artifacts directly to Hugging Face Hub.
- Async Multi-Agent Orchestration: Architected a 6-agent system (Classification, Crisis, RAG, Explainer, Wellness, Semantic) utilizing a sequential-then-parallel execution pattern via asyncio.gather, reducing pipeline latency by 45% (sub-500ms total, ~50ms core classification).
- Vector RAG & Layered Fallbacks: Built a ChromaDB vector database with all-MiniLM-L6-v2 embeddings, implementing two-tier cosine similarity thresholds (0.40 strict, 0.25 floor) and curated per-emotion fallbacks to guarantee zero empty recommendation states.
- Safety Guardrails & Streaming LLM: Engineered multi-signal crisis detection (18+ keywords + confidence rules) with strict LLM bypass to surface emergency resources (988/741741); integrated Groq API (groq/compound-mini) for real-time streaming supportive responses.
- Hybrid Explainability & Observability: Integrated a hybrid explainer (fast 200+ term lexicon attributions + model-attention SHAP values), built an exporter for 5+ Prometheus metrics (:8001), structured JSON logging (structlog), and a live Streamlit monitoring dashboard.

- Testing & CI/CD: Established a 69-test suite (74% coverage) with mocked ML dependencies, running automated matrix testing on Python 3.10 via GitHub Actions.
- Stack: PyTorch • Transformers • FastAPI • Streamlit • ChromaDB • Groq API • Prometheus • pytest • Pydantic v2

ADDITIONAL PROJECTS

E-Commerce Customer Lifetime Value Optimization | [GitHub Source](#)

- Cohort & Retention Analytics: Analyzed 393K+ transactions from a UK retailer (£7.2M revenue, 4.3K customers across 37 countries). Built RFM segmentation (6 tiers), revealing that 26% of customers drove 62% of revenue.
- Data Warehouse & Visualization: Engineered end-to-end data pipelines from raw CSVs to a PostgreSQL data warehouse, executing FP-Growth market basket analysis and building interactive Power BI dashboards with DAX metrics.

Automated ETL Pipeline with Airflow & PostgreSQL | [GitHub Source](#)

- Data Pipeline & Quality Assurance: Built an automated ETL pipeline ingesting hourly weather metrics for 5 global cities via OpenWeatherMap API into PostgreSQL, orchestrated using Apache Airflow on WSL2/Ubuntu.
- Testing & Monitoring: Implemented 25 automated data-quality checks per run and 27 pytest unit tests (100% pass rate).

WORK EXPERIENCE

Backend AI Engineering Intern | FlyRank AI | Jul 2026 – Present

- Developing backend Python microservices for early-stage AI products, transformer model integration, API schema design, and backend route optimization.

Digital Marketing & Research Analyst | GCG International LLC, Kathmandu | May 2022 – May 2023

- Executed analytics-driven digital marketing campaigns with quantitative KPI tracking and consumer behavior modeling to drive revenue decisions.

EDUCATION

- MSc Information Technology (AI/ML) | Leeds Beckett University (via The British College, Kathmandu) | Result Expected: Sep 2026
- BSc (Hons) Computing — First Class Honours | Leeds Beckett University (via The British College, Kathmandu) | 2021 – 2024

CERTIFICATIONS

- Anthropic: AI Fluency (Framework & Foundations, Claude Code 101, AI Capabilities & Limitations, Agent Skills)
- Hugging Face: Fundamentals of Agents
- Udemy: SQL for Data Analytics: Zero to Hero